

Task 2:-

Given $N = 512$

Each multiply and add pair = 2 MAC operations

a) $C_{\text{dense}} = 2N^2$

$$= 2 \times (512)^2 = 524,288 \text{ Flops}$$

b) Dense memory bytes:

We got FP32 $\Rightarrow$ 4 Bytes

$$M_{\text{dense}} = 4 \times N^2 = 4 \times (512)^2$$

$$= 2 \times 524,288 = 1,048,576 \text{ Bytes}$$

c) Sparse Compute:

$$C_{\text{dense}} \times (1-s)$$

$$C_{\text{sparse}} = 2N^2 \times (1-s)$$

$$= 524,288 \times (1-s) \text{ Flops.}$$

d) sparse memory,

CSR $\rightarrow$ stores $\rightarrow$

① Actual Non-Zero values.

Each one $\rightarrow$ 4 Bytes (4B)

$$4 \times N^2 (1-s) \text{ Bytes}$$

② The Column Index for each non-zero

$$\text{Column index} = 4 \times N^2 (1-s) \text{ Bytes}$$

③ The Row Pointer Array:

$N+1$ entries, each INT 32 = 4 Bytes

$$= 4(N+1) \text{ Bytes}$$

$$m_{\text{sprz}} = 4N^2(1-s) + 4N^2(1-s) + 4(N+1)$$

$$= 8N^2(1-s) + 4(N+1) \text{ Bytes}$$

Task 2:- Flops speedup and 2x Break even.

$$S_{\text{flops}} = \frac{C_{\text{dense}}}{C_{\text{sparse}}} = \frac{2N^2}{2N^2(1-s)} = \frac{1}{1-s}$$

For 2x speedup, $s=2$

$$\frac{1}{1-s} = 2$$

$$1-s = \frac{1}{2}$$

$$s = 0.5$$

$$s = 50\%$$

50% of Weights need to be Zero to get 2x speedup.

Task 3:- Memory Break even sparsity

$$M_{\text{sparse}} = M_{\text{dense}}$$

$$8N^2(1-s) + 4(N+1) = 4N^2$$

$$2N^2(1-s) + (N+1) = N^2$$

$$2N^2 - 2N^2S + N + 1 = N^2$$

$$-2N^2S = N^2 - 2N^2 - N - 1$$

$$S = \frac{-(N^2 + N + 1)}{-2N^2}$$

$$= \frac{1}{2} + \frac{1}{2N} + \frac{1}{2N^2}$$

$$\underline{\underline{N = 512}}$$

$$S = \frac{1}{2} + \frac{1}{2 \times 512} + \frac{1}{2 \times 512^2}$$

$$= 0.5 + 0.000977 + 0.0000019$$

$$S_{\text{break even}} \approx 0.500978 \approx 50.1\%$$

The Break even is just barely above 50%.

Beoz CSR always carries the row pointer array (513 integers) no matter how sparse the matrix is. Without that overhead it would be exactly 50%.

Task 4 find to find Speedup at $s=0.9$, $BW = 320MB/s$

Dense Time =

$$m_{dense} = 4 \times 512^2 = 1048576B$$

$$T = \frac{1048576}{320 \times 10^9}$$

$$= 3.28 \mu s$$

Sparse Bytes at $s=0.9$:-

$$\text{No. of. Non-Zeros} = N^2(1-s) = 26,214.4 \xrightarrow{(0.1)} \\ \approx 26,214$$

$$m_{sparse} = 1048576 + 1048576 + 2052 \\ = 211764B$$

$$T_{sparse} = \frac{211764}{320 \times 10^9} = 0.662 \mu s.$$

Step 3 - Speedup

$$S = \frac{T_{\text{time}}}{T_{\text{space}}}$$

$$= \frac{3.28}{0.662}$$

$$= 4.95x$$

$$\approx 5x$$

As $N \rightarrow \infty$, the row ptr turn becomes negligible and the theoretical limit is $\frac{1}{2(1-0.9)}$

$$= \frac{1}{0.2} = 5x$$

The Tiny short fall from $5x$ is caused by those STG row ptr integers, can always has to store